Quadratic Equations

Multiplying and Unmultiplying

Let's start by reviewing the facts that are usually taught to introduce quadratic equations. First, we use the distributive rule to multiply (also called FOIL):

$$(x-3)(x-4) =$$
$$x^2 - 4x - 3x + 12 =$$
$$x^2 - 7x + 12.$$

The key takeaway is that the -7 in the -7x comes from adding together -3 and -4, and the 12 comes from multiplying together -3 and -4.
Here's another example:

$$(u-3)(u+3) =$$
$$u^2 + 3u - 3u - 9 =$$
$$u^2 - 9.$$

Since we had both -3 and $+3$, the $+3$u and -3u terms canceled out, giving us a difference of squares. That will be useful later.

The reason it is useful to know what happens when multiplying is because if we can do this in reverse, we can solve quadratic equations. For example, suppose we want to find all x such that

$$x^2 - 7x + 12 = 0.$$

We already know that this is the same (has exactly the same solutions) as

$$(x-3)(x-4) = 0.$$

The only way for two numbers to multiply to zero is if one (or both) are zero. (The formal justification of this zero-product property uses the basic axiom that you can divide by any nonzero number: suppose for contradiction that $ab = 0$ with both a and b nonzero. Then by dividing both sides of the equation $ab = 0$ by a, we get $b = 0$, contradiction.)

So, the x that work are precisely those where $x - 3 = 0$ (which is $x = 3$), or $x - 4 = 0$ (which is $x = 4$). Note that the solutions are the numbers we subtract from x, i.e., not -3 and -4, but 3 and 4. Importantly, these are all the solutions.

Setting Up for Factoring

Let's try the reverse process for the example$x^2 - 2x - 24$.It would be great if we could factorize it into something like $(x -)(x -)$. Students haven't yet learned that it's always possible to find such a factorization, but our approach will also prove to them that it is always possible! By the previous section, if we managed to factorize, then whatever ends up in those blank spaces will be the solutions. But what would work in those blank spaces? Two numbers

which have sum 2 and product −24. The most commonly taught method is to find these numbers by guess and check. That can be frustrating, especially when there are negative numbers to try, and when the product has a lot of possible factorizations (24 has a ton of possibilities).

Insight: Factoring Without Guessing

Here's a way to pinpoint numbers that work without any guessing at all! The sum of two numbers is 2 when their average is 1. So, we can try to look for numbers that are 1 plus some amount, and 1 minus the same amount. All we need to do is to find if there exists a number u such that 1+ u and 1− u work as the two numbers, and u is allowed to be 0.

By looking for two numbers of the form 1+ u and 1+ u , they automatically sum to 2. So, we just need them to multiply to −24. We wish to find if there exists a u which satisfies:

$$(1 + u)(1 - u) = -24.$$

We already saw a pattern like this, where we have a sum of two numbers, multiplied by their difference. The answer is always the difference of their squares! So, by rewriting the left hand side in equivalent form, we wish to find if there exists a u such that1−u^2=−24.This is exciting! There is a lone u^2, and everything else is just a number! That means that we can finish searching for a valid u by following our nose, instead of requiring any new methods. We want:

$$u^2 = 25,$$

which we can get from $u = 5$. So, a choice for u exists! (We could alternatively have chosen $u = -5$, but that would end up giving the same result.) Therefore, tracing the logic back upward, we know that 1−5=−4 and 1+5=6 will definitely be two numbers which have sum 2 and product −24. The fact that those numbers satisfy the sum and product relations means that the factorization exists, which also means that we have found the full set of solutions:
$x = 6$ or $x = -4$.
Note that in this approach, we only need the existence of one particular number whose square equals another particular number. In this example, it is obvious that 5 is a number whose square equals 25. Once we have one such number, we can already follow through our logical steps, and we deduce a complete set of solutions to the original quadratic.
As I noticed, although I (like many others) independently came up with the trick of how to find two numbers given their sum and product, the Babylonians and Greeks already knew that particular trick thousands of years prior. However, mathematics had not been sufficiently developed for them to be able to use that trick on its own to solve general quadratic equations. Specifically, they did not work with polynomial factoring or negative numbers (not to mention non-real complex numbers).

Example of Use: a Quadratic That Can't Be Factored Easily

Now that guessing has been eliminated, we can actually solve any quadratic with this method. Consider this example:

$$\frac{x^2}{2} - 2x + 3 = 0$$

First, let's clean it up by multiplying both sides by 2, to obtain an equation whose solution set is identical:

$$x^2 - 4x + 6 = 0$$

Just like before, **if** we can find two numbers with sum 4 and product 6, then the factorization $(x -)(x -)$ will exist, and those two numbers will be the solutions. Halving the sum to get the average, we see that we'd be done if we can find some u so that numbers of the form $2 + u$ and $2 - u$ give a product of 6. These two equations are equivalent to each other:

$$4 - u^2 = 6$$
$$u^2 = -2.$$

We can satisfy the bottom equation by choosing $u = i\sqrt{2}$. Importantly, the mathematical invention of complex numbers allows us to take the square root of a negative number, so there is a valid choice for u. So, there are indeed two numbers with sum 4 and product 6, and they are $2 + u$ and $2 - u$ give, which are : $2 \pm i\sqrt{2}$. The fact that those numbers satisfy the sum and product relations means that the factorization $(x -)(x -)$ exists, and so we have found the solutions. We completed the problem, and we didn't need to use any memorized formula at all! This method works for every quadratic equation, without needing any memorization, and every step has a simple mathematical justification.